

Practice Quiz: Planning

Name: __ Date: __

Part A: Multiple Choice

1. In Active Inference, planning is: A) A completely separate cognitive system from perception B) Inference extended into the future — evaluating possible action sequences by imagining consequences C) Always conscious and deliberate D) Something only humans can do
2. Deep temporal models allow agents to: A) Only plan one step ahead B) Plan at multiple timescales simultaneously (seconds, hours, years) C) Predict the future perfectly D) Only plan for short-term goals
3. Imagination in Active Inference is: A) Disconnected from the generative model B) The same generative model running in offline simulation mode with sensory input suppressed C) Only possible in creative people D) A waste of cognitive resources
4. A habit in Active Inference is: A) An action the agent always regrets B) A policy with a strong prior that is selected without explicit EFE evaluation C) A random action repeated by coincidence D) Only possible after formal training
5. The horizon problem refers to: A) The difficulty of seeing far away B) The fact that planning further ahead requires considering exponentially more possibilities C) The curvature of the earth limiting visibility D) A computational error in simulations
6. Bounded rationality means agents are: A) Irrational most of the time B) Rational given their computational and informational constraints C) Only rational about simple decisions D) Unable to learn from experience
7. Breaking a habit requires: A) Simply deciding to stop B) Overriding the strong prior on the habitual policy with deliberate planning and new prediction errors C) Complete memory loss D) External punishment

Part B: Short Answer

1. A student must decide between studying for an exam tomorrow (certain, immediate benefit) and working on a semester-long research project (uncertain, long-term benefit). Use deep temporal models and Expected Free Energy to analyze this decision. What factors influence which option the student chooses?
2. Explain why planning and memory might use the same neural systems. How does Active Inference predict that remembering the past and imagining the future are related processes?
3. Reflect on the entire 8-module sequence (Systems → Agents → Perception → Cognition → Action → Learning → Communication → Planning). Choose any two modules and explain how they depend on each other. Why couldn't we have taught them in the opposite order?